

Claude Code CLI — EntraID JWT Authentication via AIGW

Configures the Claude Code CLI to send requests through this self-hosted AIGW gateway, authenticated using a Microsoft EntraID JWT bearer token obtained automatically from the local Azure CLI session.

How it works

Claude Code CLI supports an `apiKeyHelper` — a script it calls on each session start whose stdout becomes the bearer token sent to the gateway. `get-az-token.sh` in this repo fills that role: it fetches an EntraID access token from the local Azure CLI cache, triggering an interactive browser login if the session has expired.

```
Claude Code CLI
├── calls get-az-token.sh → outputs EntraID access token
└── sends requests to AIGW gateway (http://localhost:8787)
    └── Authorization: Bearer <token>
        └── gateway validates JWT, maps claims → Portkey workspace
```

Prerequisites

- The gateway stack running
- The EntraID app registration configured for this gateway — see [entraid-jwt-auth guide](#) or run `Setup-EntraID-App.sh`
- `ENTRAID_CLIENT_ID` and `ENTRAID_TENANT_ID` available — set them in the `env` block of `settings.json` (step 2), or let `Setup-EntraID-App.sh` write them for you
- Azure CLI installed (`brew install azure-cli`) and logged in at least once

1. Install the token helper script

Copy `get-az-token.sh` from this repo to your Claude config directory:

```
cp get-az-token.sh ~/.claude/get-az-token.sh
chmod +x ~/.claude/get-az-token.sh
```

The script reads `ENTRAID_CLIENT_ID` and `ENTRAID_TENANT_ID` from the environment. Supply them via the `env` block in Claude Code's settings (step 2) — that is the recommended approach and no other configuration file is needed.

2. Configure Claude Code CLI

Edit (or create) `~/.claude/settings.json`:

```
{
  "apiKeyHelper": "/Users/<you>/.claude/get-az-token.sh",
  "env": {
    "ANTHROPIC_BASE_URL": "http://localhost:8787",
    "ENTRAID_CLIENT_ID": "<your-app-client-id>",
    "ENTRAID_TENANT_ID": "<your-tenant-id>",
    "ANTHROPIC_DEFAULT_SONNET_MODEL": "anthropic.claude-sonnet-5",
    "ANTHROPIC_DEFAULT_OPUS_MODEL": "anthropic.claude-opus-4-8",
    "ANTHROPIC_DEFAULT_HAIKU_MODEL": "anthropic.claude-haiku-4-5-20251001",
    "CLAUDE_CODE_API_KEY_HELPER_TTL_MS": "3000000"
  }
}
```

Field	Value
apiKeyHelper	Absolute path to <code>get-az-token.sh</code>
ANTHROPIC_BASE_URL	Your gateway URL — use <code>http://localhost:8787</code> for local Docker
ENTRAID_CLIENT_ID	The Application (client) ID from the EntraID app registration
ENTRAID_TENANT_ID	Your Entra tenant ID
ENTRAID_RESOURCE_URI	(optional) App ID URI to request a token for — defaults to <code>api://<ENTRAID_CLIENT_ID></code> . Override only if your app registration uses a custom URI.
ANTHROPIC_DEFAULT_SONNET_MODEL	Model ID Claude Code uses for Sonnet requests — must include the <code>anthropic.</code> prefix when routing via Vertex AI
ANTHROPIC_DEFAULT_OPUS_MODEL	Model ID for Opus requests — same prefix requirement
ANTHROPIC_DEFAULT_HAIKU_MODEL	Model ID for Haiku requests — same prefix requirement
CLAUDE_CODE_API_KEY_HELPER_TTL_MS	How long (ms) Claude Code caches the token from <code>apiKeyHelper</code> — set to <code>3000000</code> (~50 min) to match the Azure CLI token lifetime and avoid unnecessary re-fetches

Tip: If you have `ENTRAID_CLIENT_ID` and `ENTRAID_TENANT_ID` in your shell environment already (e.g. via `direnv`), you can omit those two from the `env` block and the script will pick them up directly. The model vars should always be set explicitly when using this config.

Project-level config (recommended when mixing auth methods)

If you use a different Claude Code config for other repos (e.g. a direct Portkey cloud setup), put the gateway settings in a **project-level** `.claude/settings.json` at the root of this repo instead of the global `~/.claude/settings.json`. Claude Code merges the two — project settings override global ones for the same keys — so your global config stays intact for everything else:

```
mkdir -p .claude
# create .claude/settings.json with only the gateway-specific overrides
```

```
{
  "apiKeyHelper": "/Users/<you>/.claude/get-az-token.sh",
  "env": {
    "ANTHROPIC_BASE_URL": "http://localhost:8787",
    "ENTRAID_CLIENT_ID": "<your-app-client-id>",
    "ENTRAID_TENANT_ID": "<your-tenant-id>",
    "ANTHROPIC_CUSTOM_HEADERS": "x-portkey-config: <your-config-id>",
    "ANTHROPIC_DEFAULT_SONNET_MODEL": "anthropic.claude-sonnet-5",
    "ANTHROPIC_DEFAULT_OPUS_MODEL": "anthropic.claude-opus-4-8",
    "ANTHROPIC_DEFAULT_HAIKU_MODEL": "anthropic.claude-haiku-4-5-20251001",
    "CLAUDE_CODE_API_KEY_HELPER_TTL_MS": "3000000"
  }
}
```

Theme, permissions, and any other global settings are inherited automatically.

3. Set the Portkey config header (optional)

If your gateway requires an explicit routing config (the `x-portkey-config` header), the recommended approach is to configure a **default config** in the AIGW control plane for your workspace — the gateway then applies it to all requests that don't carry the header explicitly.

Alternatively, you can set it per-project in `.claude/settings.json` at the repo level if Claude Code CLI adds support for custom request headers in your version.

4. Verify with curl

Before starting Claude Code, confirm the token and gateway are working end-to-end:

```
TOKEN=$(./get-az-token.sh) && \
curl http://127.0.0.1:8787/v1/messages \
  -H "x-portkey-config: <your-config-id>" \
  -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d '{"model": "anthropic.claude-sonnet-5", "max_tokens": 250, "messages": \
[{"role": "user", "content": "hi"}]}'
```

Model prefix: when routing via Vertex AI, model names must be prefixed with `anthropic.` (e.g. `anthropic.claude-sonnet-5`). Without the prefix the gateway returns `messages is not supported by vertex-ai`.

If the request succeeds, inspect the token claims to verify the gateway claims are correct:

```
echo $TOKEN | cut -d. -f2 | base64 -d 2>/dev/null | python3 -m json.tool
```

Check that `portkey_oid`, `email_id`, and `uid` are present and match your deployment's `ORGANISATIONS_TO_SYNC` value.

5. Start Claude Code

```
claude
```

On the first run (or after your Azure CLI session expires) a browser window opens to sign in to EntraID. After authentication the session is cached by the Azure CLI — subsequent invocations are silent until the token expires.

To confirm requests are landing in the gateway, check the control plane logs or AIGW analytics for your workspace.

Troubleshooting

Symptom	Likely cause	Fix
AADSTS650057: Invalid resource	App not exposed as an API, or Azure CLI not pre-authorized	Run the fix commands below, or re-run <code>Setup-EntraID-App.sh</code> on the existing app
AADSTS501461: AcceptMappedClaims...	Resource was requested as <code>api://<id></code> instead of bare GUID	<code>ENTRAID_RESOURCE_URI</code> must be the bare client GUID, not the <code>api://</code> URI — check the <code>env</code> block in <code>settings.json</code>
AADSTS65001: consent required	Azure CLI not pre-authorized on the resource app	Pre-authorize Azure CLI (see fix commands below)
ENTRAID_CLIENT_ID not set	Variable not in the <code>env</code> block	Add it to the <code>env</code> block in <code>settings.json</code>
messages is not supported by vertex-ai	Model name missing	Use <code>anthropic.claude-sonnet-5</code> not <code>claude-sonnet-5</code> when targeting Vertex AI

Symptom	Likely cause	Fix
	<code>anthropic.</code> prefix	
API errors after copying entra config to global <code>settings.json</code>	Model env vars absent — Claude Code defaults to bare model names without the <code>anthropic.</code> prefix	Add <code>ANTHROPIC_DEFAULT_SONNET_MODEL</code> , <code>ANTHROPIC_DEFAULT_OPUS_MODEL</code> , and <code>ANTHROPIC_DEFAULT_HAIKU_MODEL</code> with the <code>anthropic.</code> prefix to the <code>env</code> block
<code>401 Unauthorized</code> from gateway	JWT invalid or missing claims	Decode the token (see below) and verify <code>portkey_oid</code> and <code>email_id</code> are present
Invalid API Key (Error Code: 03)	Bearer token is a literal string, not an actual token	Use <code>\$(./get-az-token.sh)</code> not <code>{./get-az-token.sh}</code> for command substitution
Browser opens on every run	Azure CLI token cache expired	Normal — sign in; the cache is then reused for the token lifetime (~1 h)
<code>az: command not found</code>	Azure CLI not installed	<code>brew install azure-cli</code>
Gateway returns <code>502</code> / no route	No Portkey config header and no default config set	Set a default config in the AIGW control plane for your workspace

Decode the token to inspect claims:

```
echo $TOKEN | cut -d. -f2 | base64 -d 2>/dev/null | python3 -m json.tool
```

Fixing AADSTS650057 on an existing app registration

If the app was created before the CLI token flow was needed, expose the API manually:

```
APP_OBJECT_ID=$(az ad app show --id "$ENTRAID_CLIENT_ID" --query "id" -o tsv)
SCOPE_ID=$(python3 -c "import uuid; print(uuid.uuid4())")

az rest --method PATCH \
  --uri "https://graph.microsoft.com/v1.0/applications/$APP_OBJECT_ID" \
  --headers "Content-Type=application/json" \
  --body "{
    \"identifierUri\": [\"api://$ENTRAID_CLIENT_ID\"],
    \"api\": {
      \"oauth2PermissionScopes\": [{
        \"id\": \"$SCOPE_ID\",
```

```
    \"adminConsentDescription\": \"Access the AIGW gateway\",
    \"adminConsentDisplayName\": \"Access AIGW gateway\",
    \"isEnabled\": true,
    \"type\": \"User\",
    \"value\": \"user_impersonation\"
  }
}
```

After running this, `az account get-access-token --resource \"api://$ENTRAID_CLIENT_ID\"` will succeed.

Updating credentials

If you rotate the EntraID app registration or move to a different tenant, update `ENTRAID_CLIENT_ID` / `ENTRAID_TENANT_ID` in:

1. `~/.claude/settings.json` (env block) — or the project-level `.claude/settings.json` if you used that approach
2. Re-run `az login --tenant <new-tenant-id>` to refresh the CLI session